

Composing layers

Project: Image-pattern Network Investigation

1. Your mission

Your learning target

I can trace how one layer supplies the next layer inputs.

I can check that connected layer dimensions and feature order agree.

I can explain why intermediate activations are not automatically class predictions.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Activation functions. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Dot products and weighted sums. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

This week you will practise:

Trace how one layer supplies the next layer inputs

Check that connected layer dimensions and feature order agree

Explain why intermediate activations are not automatically class predictions

Retrieve what you already know

For $z = -2$ and $z = 3$, what does ReLU return? Does this operation itself assign a class?

My starting explanation

This week's end goal

By the end, I can explain composing layers using a traced example and a stated limitation.

2. Notice the evidence

A network connects layers so that the output vector of one layer becomes the input vector of the next. Use a fictional numerical input $x = [x_1, x_2]$. The first layer computes $z_1 = x_1 - x_2$ and $z_2 = x_1 + x_2 - 2$, then applies ReLU separately: $h_1 = \max(0, z_1)$, $h_2 = \max(0, z_2)$. Its output is the ordered vector $h = [h_1, h_2]$.

The next layer computes $y = 2h_1 - h_2 + 1$. It receives h , not the original x . It uses weights $[2, -1]$ and bias 1, with identity as its output activation. This is a two-layer computation when counting the layers with weights; an input placeholder is not counted here. All parameters are supplied, so this lesson performs no training.

Stage · Input size · Output size

Stage: First layer with ReLU

Input size: 2 components

Output size: 2 components

Stage: Output layer with identity

Input size: 2 components

Output size: 1 component

The two first-layer units both receive the same original x . They can be calculated in either order. Only after both activations are available do we evaluate the output layer.

List the given facts and one fact that is not supplied.

3. See how it works

Take $x = [3, 1]$. First compute $z_1 = 3 - 1 = 2$ and $z_2 = 3 + 1 - 2 = 2$. ReLU gives $h = [2, 2]$. The output layer receives exactly this ordered pair and returns $y = 2 \text{ times } 2 - 2 + 1 = 3$. Record z , h and y separately so that a correct final number cannot hide an incorrect intermediate step.

```
def forward(x):
    if len(x) != 2:
        raise ValueError("Expected two input components")

    x1, x2 = x

    z = [x1 - x2, x1 + x2 - 2]

    h = [max(0, value) for value in z]

    y = 2 * h[0] - h[1] + 1

    return z, h, y

print(forward([3, 1])) # ([2, 2], [2, 2], 3)
print(forward([1, 3])) # ([-2, 2], [0, 2], -1)
```

The code assumes numeric inputs with the agreed meaning. Its length check does not validate units or field order. Python indices 0 and 1 correspond to the first and second hidden components. There is no feedback from y into h during this forward calculation.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

Compare a familiar case with a changed case

A revised hidden layer returns [h1, h2, h3]. The output layer still has weights [2, -1]. Explain the mismatch and what must be specified before calculation.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change: Change only v from 3 to 1.

Prediction: h1 stays 2, h2 becomes 0, y becomes 2.

Reason: The second hidden unit's preactivation reaches 0 while the first unit is unchanged.

5. Words that unlock the idea

Word · Meaning

Word: Layer

Meaning: A group of computations that maps an input vector to an output vector

Word: Hidden activation

Meaning: An intermediate value computed inside a network

Word: Composition

Meaning: Applying one function to the output of another

Word: Forward calculation

Meaning: Computing successive values from an input using fixed parameters

Use two terms to explain a step in this week's example.

6. Follow a worked example

Complete each stage before using its output

Available input · Computation · Value passed on

Available input: x is $[3, 1]$

Computation: Both first layer scores

Value passed on: z is $[2, 2]$

Available input: z is $[2, 2]$

Computation: ReLU on each component

Value passed on: h is $[2, 2]$

Available input: h is $[2, 2]$

Computation: $2h_1 - h_2 + 1$

Value passed on: y is 3

Read the rows from top to bottom. The first two rows together form the hidden layer; the final row is the output layer. Each row shows the values actually passed to the next stage.

Trace $x = [1, 1]$. Record z_1 , z_2 , h_1 , h_2 and y . Explain why using x directly in the output formula would be a different model.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Composing layers

Compose two hidden detectors before one output instead of skipping a layer.

The facts we will use

For input $[u, v]$, $h_1 = \text{ReLU}(u-2)$, $h_2 = \text{ReLU}(v-1)$, $y = h_1 - h_2$. Input $[4, 3]$.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Composing layers

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. First hidden unit

$u-2=2$, so $h1=2$.



2. Second hidden unit

$v-1=2$, so $h2=2$.



3. Compose output

$y=2-2=0$; the output uses activated hidden values, not raw inputs.

CHANGE ONE THING

Change only v from 3 to 1.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. First hidden unit

What happens and why: $u-2=2$, so $h1=2$.

Step: 2. Second hidden unit

What happens and why: $v-1=2$, so $h2=2$.

Step: 3. Compose output

What happens and why: $y=2-2=0$; the output uses activated hidden values, not raw inputs.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Composition means one layer's result becomes the next layer's input. Compute both hidden units before combining them. At $[4,3]$, their equal outputs cancel. Lowering only the second input removes its positive hidden contribution, so subtracting it no longer reduces the output. The final subtraction must use activated hidden values rather than the raw measurements.

What this example does not establish

Hand-composed detectors illustrate layer order; they are not evidence of a learned or biologically realistic neural mechanism.

Read the labelled picture: What intermediate value explains the changed output? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Change one thing: Change only v from 3 to 1. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

The output layer expects two inputs because its two weights match h_1 and h_2 . If the hidden layer is redesigned to return three activations, the output layer needs a compatible three-input definition. Silently dropping an activation or inventing a third weight does not preserve the stated model.

Order matters even when lengths match. For $x = [4, 1]$, $h = [3, 3]$ happens to be unchanged by swapping, so that case cannot reveal an order bug. For $x = [3, 0]$, $h = [3, 1]$ gives $y = 6$. Swapping only those hidden values gives $2 \text{ times } 1 - 3 + 1 = 0$. Swapping both the values and their matching weights would preserve the dot product.

Keeping the output layer fixed, replacing ReLU with identity changes the model. For $x = [1, 3]$, identity gives $h = [-2, 2]$ and $y = -5$; ReLU gives $h = [0, 2]$ and $y = -1$. More layers or a different activation do not establish better predictions without valid evaluation.

Locate the earliest unsupported step

Compare hidden vectors for $x = [4, 1]$ and $x = [3, 0]$. Which input reveals a bug that swaps h_1 and h_2 but keeps output weights unchanged? Give correct and swapped outputs for that input.

State one repair and the checks you would rerun.

8. Practise together

For $x = [1, 3]$, write both scores, both ReLU activations and y using the lesson model.

Trace $x = [1, 1]$. Record $z1, z2, h1, h2$ and y . Explain why using x directly in the output formula would be a different model.

Explain your prediction to a partner before comparing results.

Practise with less help: Repair a dimension mismatch

Use workbook W2 for the connected example “Composing layers”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Composing layers. Proposed explanation: “The second hidden unit receives the first hidden unit output”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

What intermediate value explains the changed output? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

Predict, investigate and preserve the evidence

A revised hidden layer returns [h1, h2, h3]. The output layer still has weights [2, -1]. Explain the mismatch and what must be specified before calculation.

```
def forward(x):  
    if len(x) != 2:  
        raise ValueError("Expected two input components")  
  
    x1, x2 = x  
    z = [x1 - x2, x1 + x2 - 2]  
    h = [max(0, value) for value in z]  
    y = 2 * h[0] - h[1] + 1  
    return z, h, y  
  
print(forward([3, 1])) # ([2, 2], [2, 2], 3)
```

```
print(forward([1, 3])) # ([-2, 2], [0, 2], -1)
```

Record · Your evidence

Record: Given input and fixed rule

Your evidence: _____

Record: Prediction before checking

Your evidence: _____

Record: Actual result, or “not run”

Your evidence: _____

Record: First mismatch and one explanation

Your evidence: _____

Record: Boundary or missing input and expected response

Your evidence: _____

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

Explain → apply → challenge

For $x = [1, 3]$, write both scores, both ReLU activations and y using the lesson model.

Compare hidden vectors for $x = [4, 1]$ and $x = [3, 0]$. Which input reveals a bug that swaps h_1 and h_2 but keeps output weights unchanged? Give correct and swapped outputs for that input.

Use the lesson model to verify h and y for $[0, 1]$ and $[0, 2]$. Could a fixed later layer receiving only h distinguish the two original inputs?

Explain which evidence supports your conclusion and which claim remains unproved.

11. Connect your project

Add a labelled composing layers evidence sheet to Image-pattern Network Investigation. Show how you can trace how one layer supplies the next layer inputs. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence · Record

Project evidence: Component and version

Record: _____

Project evidence: Inputs permitted at prediction time

Record: _____

Project evidence: Trace and comparison with earlier work

Record: _____

Project evidence: Failure, limit and human review

Record: _____

Evidence for your project

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

The computed y is a numerical output, not a verified probability or a class by itself. If we separately declare class 1 when $y \geq 0$ and class 0 otherwise, the worked inputs $[3, 1]$ and $[1, 3]$ receive classes 1 and 0. Their true labels are not supplied, so these calculations do not measure accuracy.

A hidden value is a computed feature. We should not name h_1 as confidence or h_2 as danger without evidence for that interpretation. Distinct inputs may also produce the same hidden vector: $[0, 1]$ and $[0, 2]$ both give $h = [0, 0]$ and $y = 1$. A later deterministic layer seeing only h cannot recover the lost distinction.

A team says its output is correct because all layers connect and its code runs. Which claim has it checked, and what evidence is still missing?

Next week uses longer forward-pass tables to locate the first incorrect intermediate value and check complete network calculations.

Use the lesson model to verify h and y for $[0, 1]$ and $[0, 2]$. Could a fixed later layer receiving only h distinguish the two original inputs?

My limited claim and a useful next check

13. Study a complete case

Complete evidence record

Complete each stage before using its output

Available input · Computation · Value passed on

Available input: x is $[3, 1]$

Computation: Both first layer scores

Value passed on: z is [2, 2]

Available input: z is [2, 2]

Computation: ReLU on each component

Value passed on: h is [2, 2]

Available input: h is [2, 2]

Computation: $2h_1 - h_2 + 1$

Value passed on: y is 3

Read the rows from top to bottom. The first two rows together form the hidden layer; the final row is the output layer. Each row shows the values actually passed to the next stage.

Follow the input through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

14. Try a new case independently

Independent case: Compose a new network independently

A new model takes $[u, v]$. Compute $a = \text{ReLU}(u + v - 3)$, $b = \text{ReLU}(u - v)$, then $q = a + 2b - 1$. Trace both inputs [2, 1] and [1, 2], including scores before ReLU. Apply class 1 when $q \geq 0$. State the hidden and output vector sizes.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

A new model takes $[u, v]$. Compute $a = \text{ReLU}(u + v - 3)$, $b = \text{ReLU}(u - v)$, then $q = a + 2b - 1$. Trace both inputs $[2, 1]$ and $[1, 2]$, including scores before ReLU. Apply class 1 when $q \geq 0$. State the hidden and output vector sizes.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: Compose two hidden detectors before one output instead of skipping a layer.

By the end, I can · How I will show it

By the end, I can: Trace how one layer supplies the next layer inputs

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Check that connected layer dimensions and feature order agree

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Explain why intermediate activations are not automatically class predictions

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Trace how one layer supplies the next layer inputs

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.